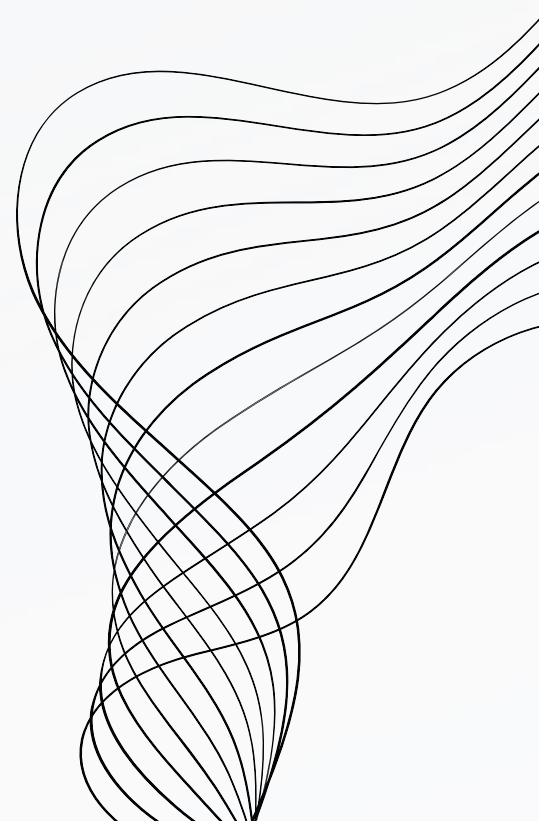


DATAVIS

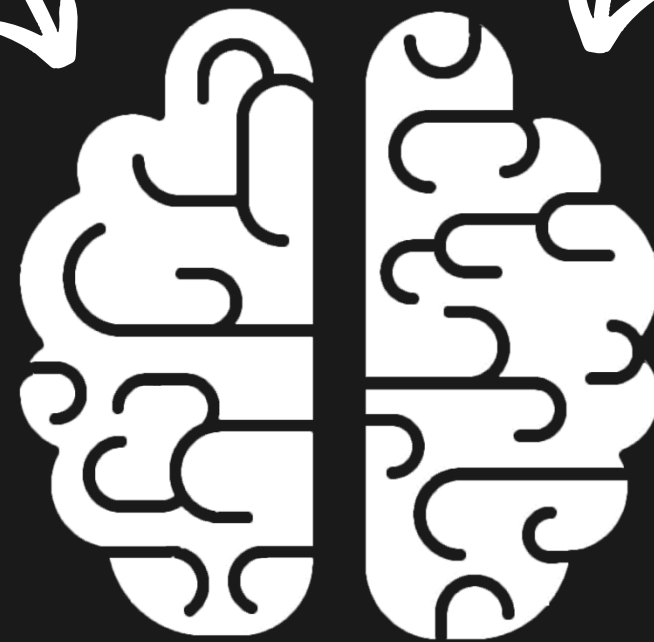
LECTURE №2. PERCEPTION. SENSATIONS. COLOR.



PERCEPTION AND UNDERSTANDING

Perception is the physical transmission of visual, audio or tactile impulses to the brain.

Understanding is the process of processing received information and drawing conclusions.



QUOTE OF THE LECTURE

Edward Tufte

«...avoiding disaster becomes the first principle of coloring information: first, do no harm.».



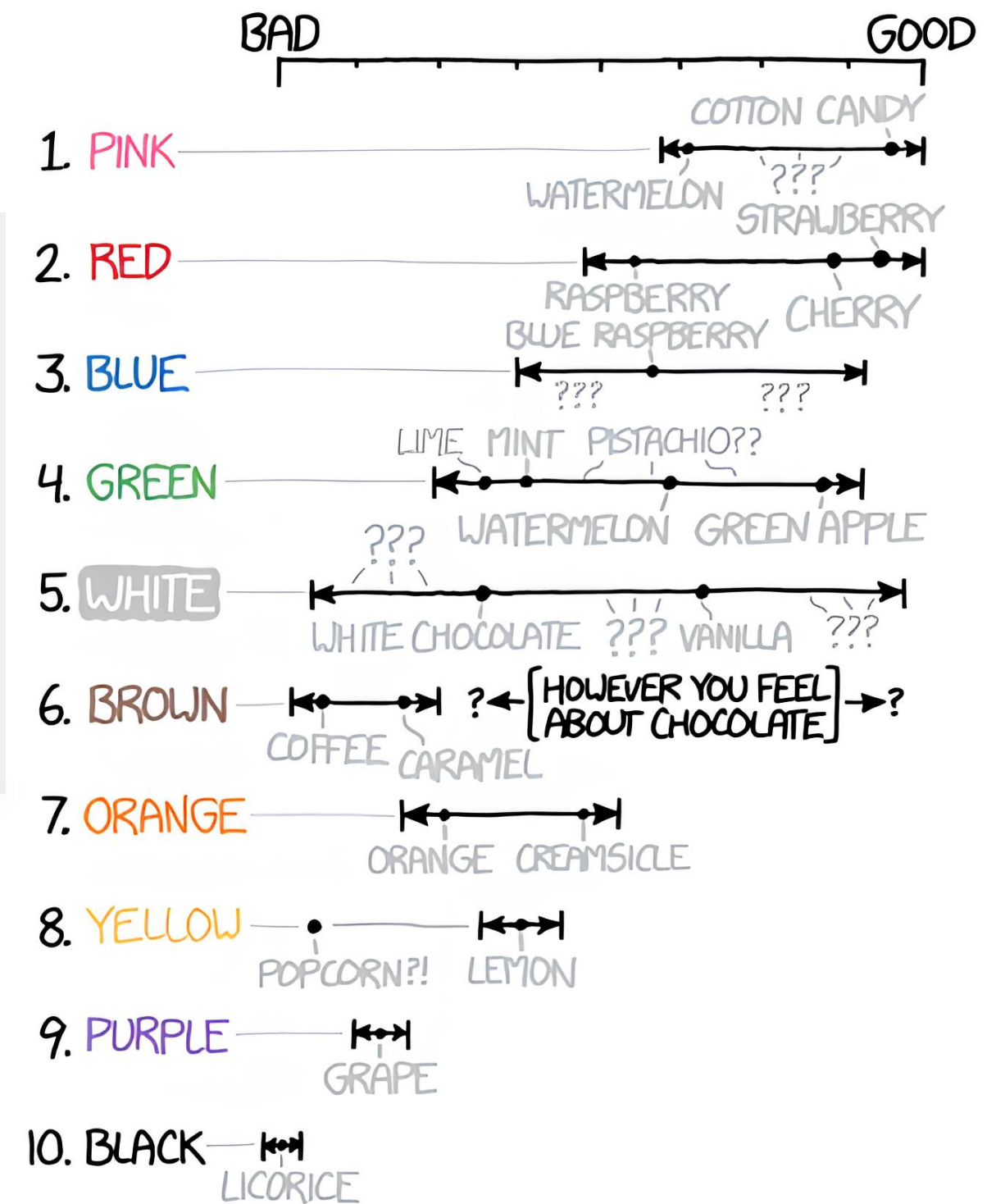
Professor Emeritus of Statistics, Political Science, and
Computer Science at Yale University

COLOR

Color is not just decoration, it is a powerful tool that can influence our perception of information.

It evokes emotions, creates associations and directs our attention.

BEST-TASTING COLORS



HOW TO CHOOSE THE RIGHT COLORS?



. A well-chosen color scheme helps to convey mood, emphasize certain design elements and create a holistic composition.

COLOR PSYCHOLOGY



Color is not just a shade, it is a powerful tool that can shape our mood and even decisions..

COLOR SCHEMES



Accessibility is an approach that takes into account the diversity of user needs and makes data more usable for everyone.

ACCESSIBILITY

COLOR PSYCHOLOGY

Color names if
you're a girl...

Maraschino		Red
Cayenne		
Maroon		Purple
Plum		
Eggplant		
Grape		
Orchid		
Lavender		
Carnation		Pink
Strawberry		
Bubblegum		
Magenta		
Salmon		
Tangerine		Orange
Cantaloupe		
Banana		Yellow
Lemon		
Honeydew		Green
Lime		
Spring		
Clover		
Fern		
Moss		
Flora		
Sea Foam		
Spindrift		Blue
Teal		
Sky		
Turquoise		

Color names if
you're a guy...

Doghouse Diaries
"We take no as an answer."

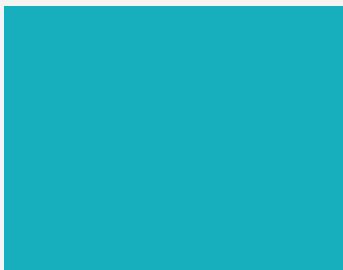
by Randall Munroe, xkcd.com

COLOR PSYCHOLOGY



PANTONE
BLUE

- Calm,
- Security,
- Integrity,
- Trust,
- Intelligence



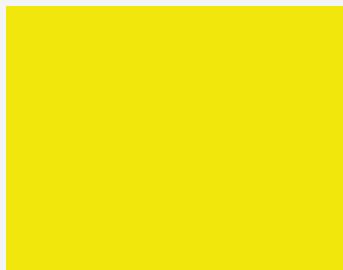
PANTONE
TURQUOISE

- Spirituality
- ,Healing
- ,Protection,
- Sophistication



PANTONE
GREEN

- Freshness,
- Nature
- ,Renewal,
- Money,
- Earth



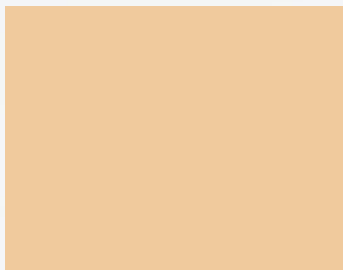
PANTONE
YELLOW

- Sun,
- Energy
- ,Warmth,
- Happiness
- ,Joy



PANTONE
BROWN

- Friendliness
- ,Earth,
- Conservativeness
- ,Longevity



PANTONE
TAN

- Dependency,
- Flexibility
- ,Value,
- Fragility



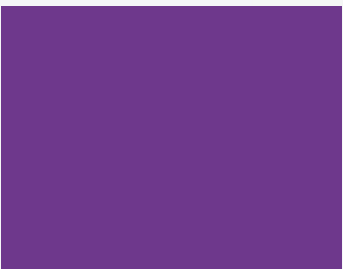
PANTONE
GOLD

- Welfare
- ,Prosperity,
- Traditionalism
- ,Value



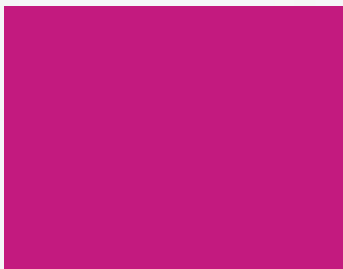
PANTONE
SILVER

- Chic
- ,High-tech,
- Elegance
- ,Smoothness



PANTONE
PURPLE

- Greatness,
- Nobility
- ,Luxury,
- Ambition



PANTONE
PINK

- Health
- ,Happiness,
- Femininity,
- Sweetness
- ,Playfulness



PANTONE
RED

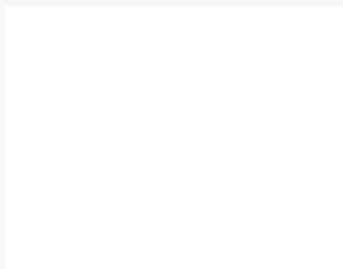
- Love
- ,Passion,
- Strength,
- Energy,
- Heat



PANTONE
ORANGE

- Courage
- ,Success,
- Friendliness
- ,Confidence

Research on colors:
Elliot A. J. (2015).
Color and
psychological
functioning: a review
of theoretical and
empirical work.
*Frontiers in
psychology, 6, 368.*



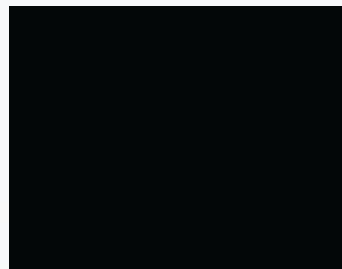
PANTONE
BLUE

- Good,
- Purity,
- Freshness
- ,Simplicity



PANTONE
GRAY

- Security,
- Intelligence
- ,Integrity
- ,Reliability



PANTONE
BLACK

- Protection,
- Elegance
- ,Drama,
- Classicism,
- Formalism

COLOR SCHEMES



Complementary



Analog



Triad



Complementary
Triad



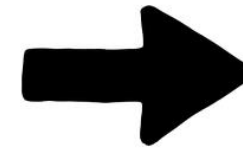
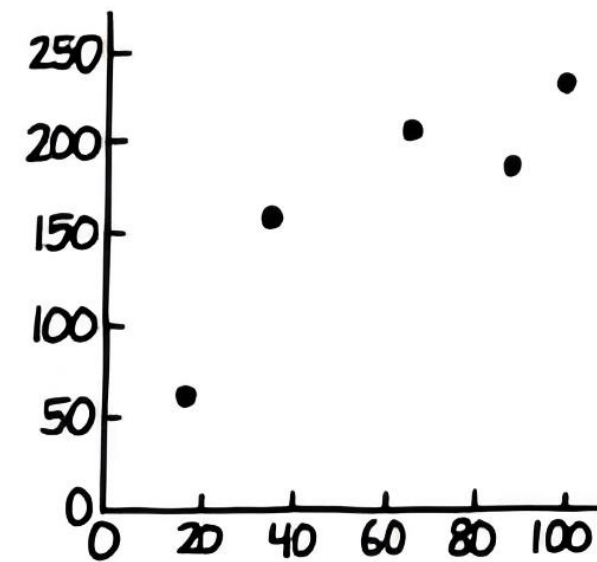
Double split



Tetrad

ACCESSIBILITY

VARIABLE 1: X AXIS
VARIABLE 2: Y AXIS

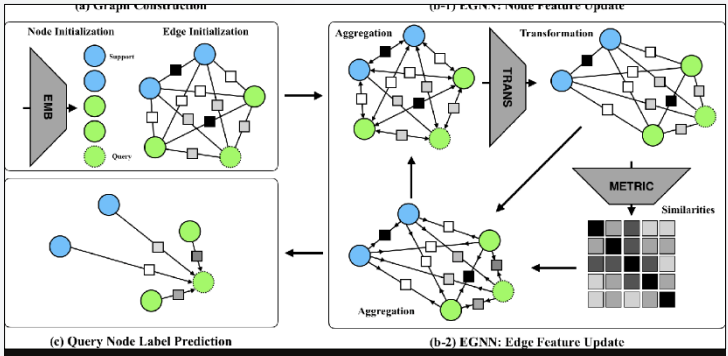


VARIABLE 1: HUE
VARIABLE 2: LABEL

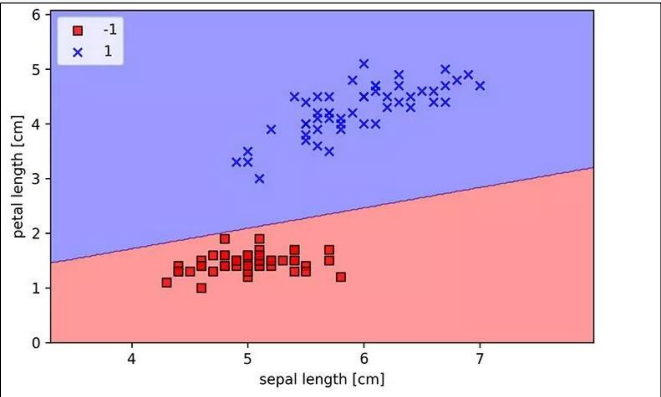
280

DESIGN TIP: YOU CAN MAKE YOUR GRAPHS MORE SPACE-EFFICIENT BY USING HUE AND LABEL FOR THE FIRST TWO VARIABLES, INSTEAD OF ONLY TURNING TO THEM ONCE YOU'VE USED UP THE X AND Y AXES.

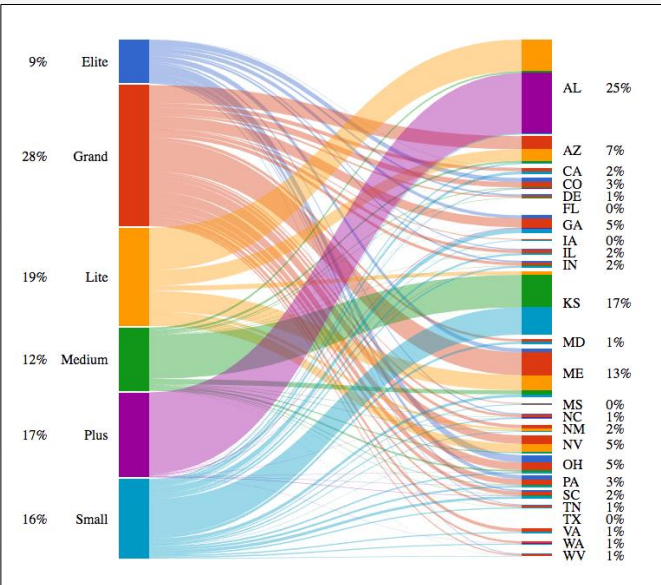
ACCESSIBILITY



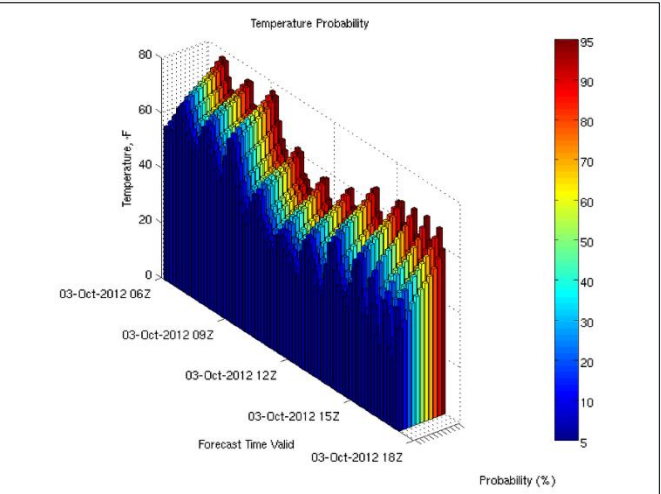
Separation of
types



Comparison
of objects



Improving
readability



Comparison of
properties

BASIC ERRORS

01

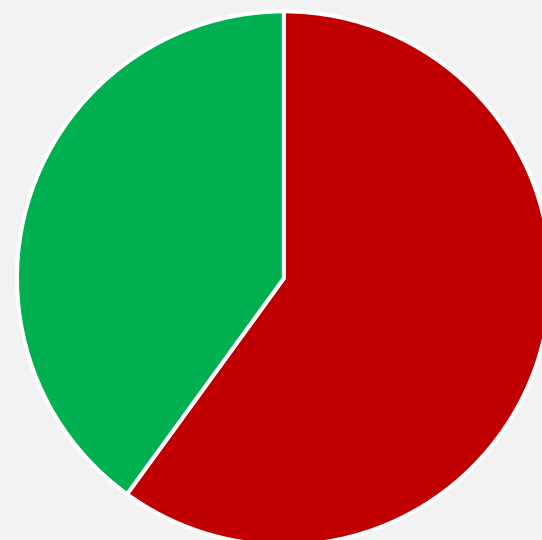
WEAK CONTRAST

02

OVERSUFFERING COLORS

03

COLOR AND DATA INCONSISTENCY



■ Yes
■ No

BASIC ERRORS

01

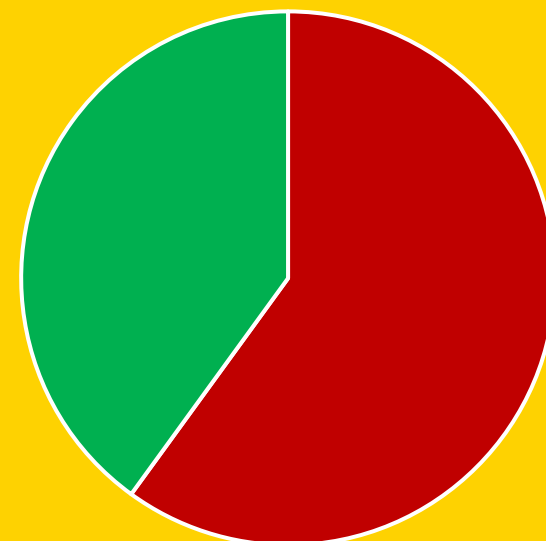
WEAK CONTRAST

02

OVERSUFFERING COLORS

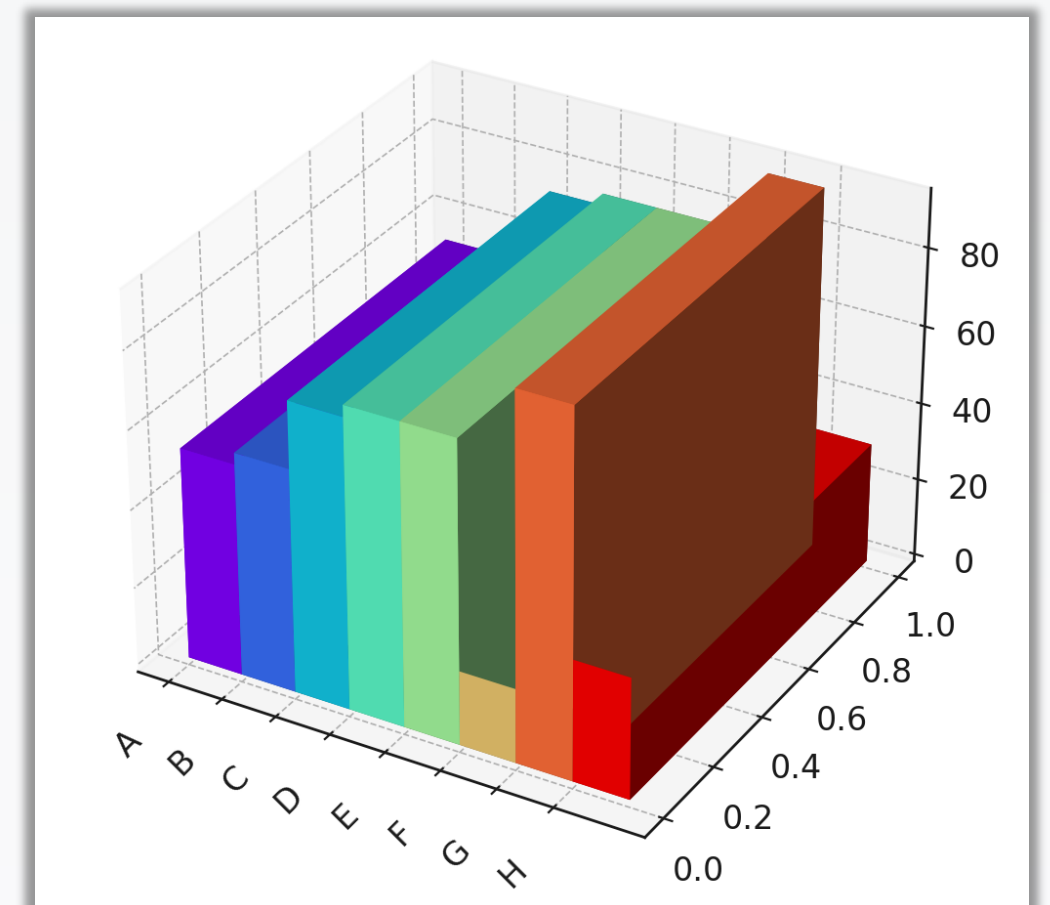
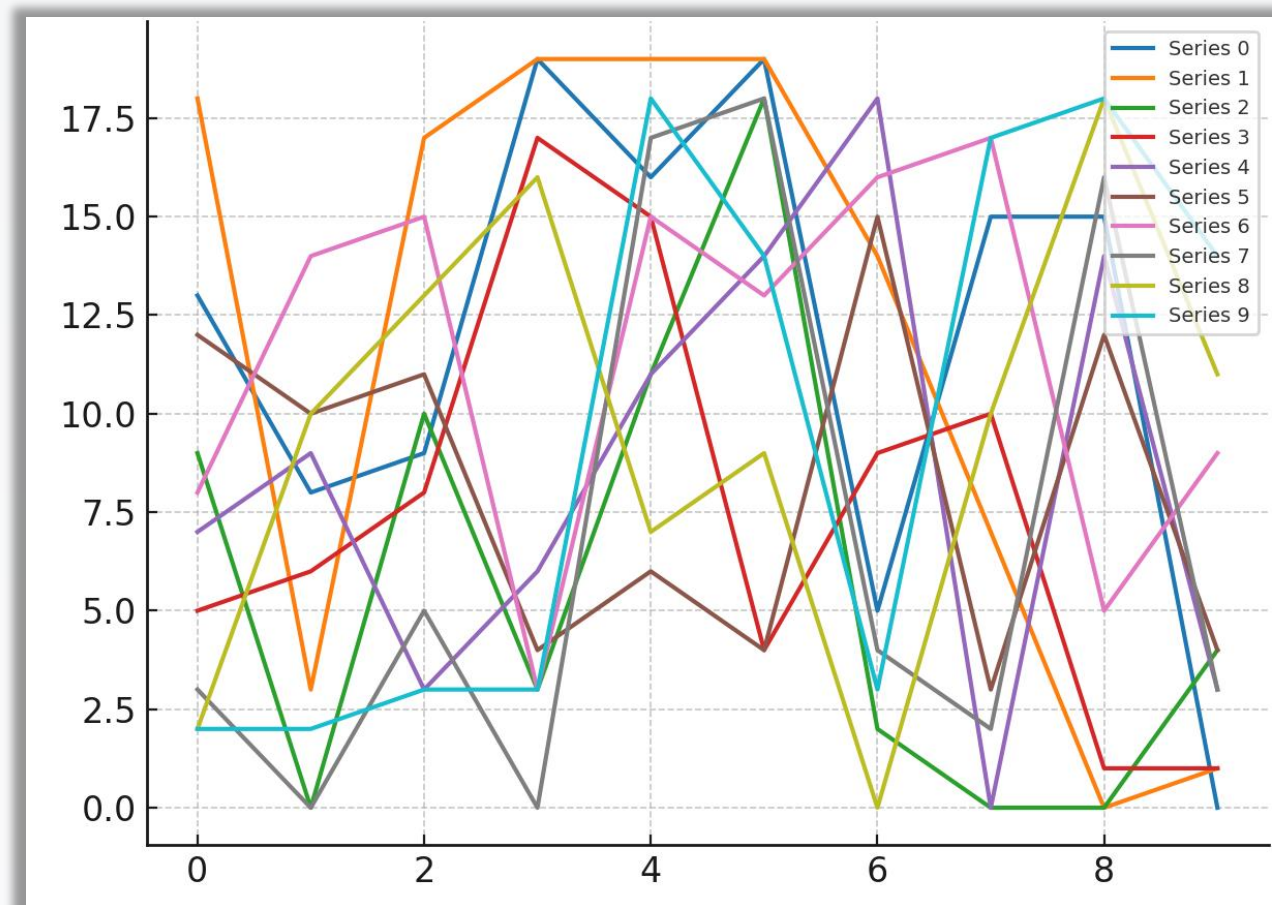
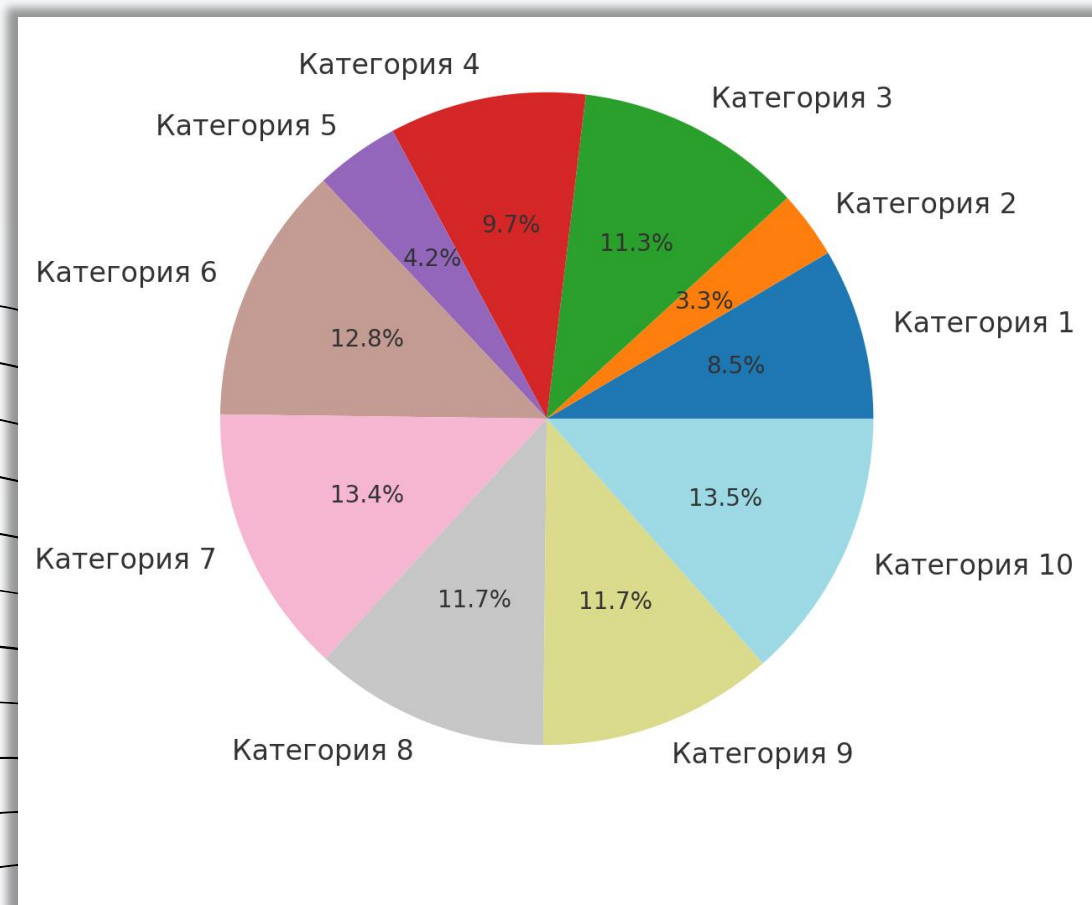
03

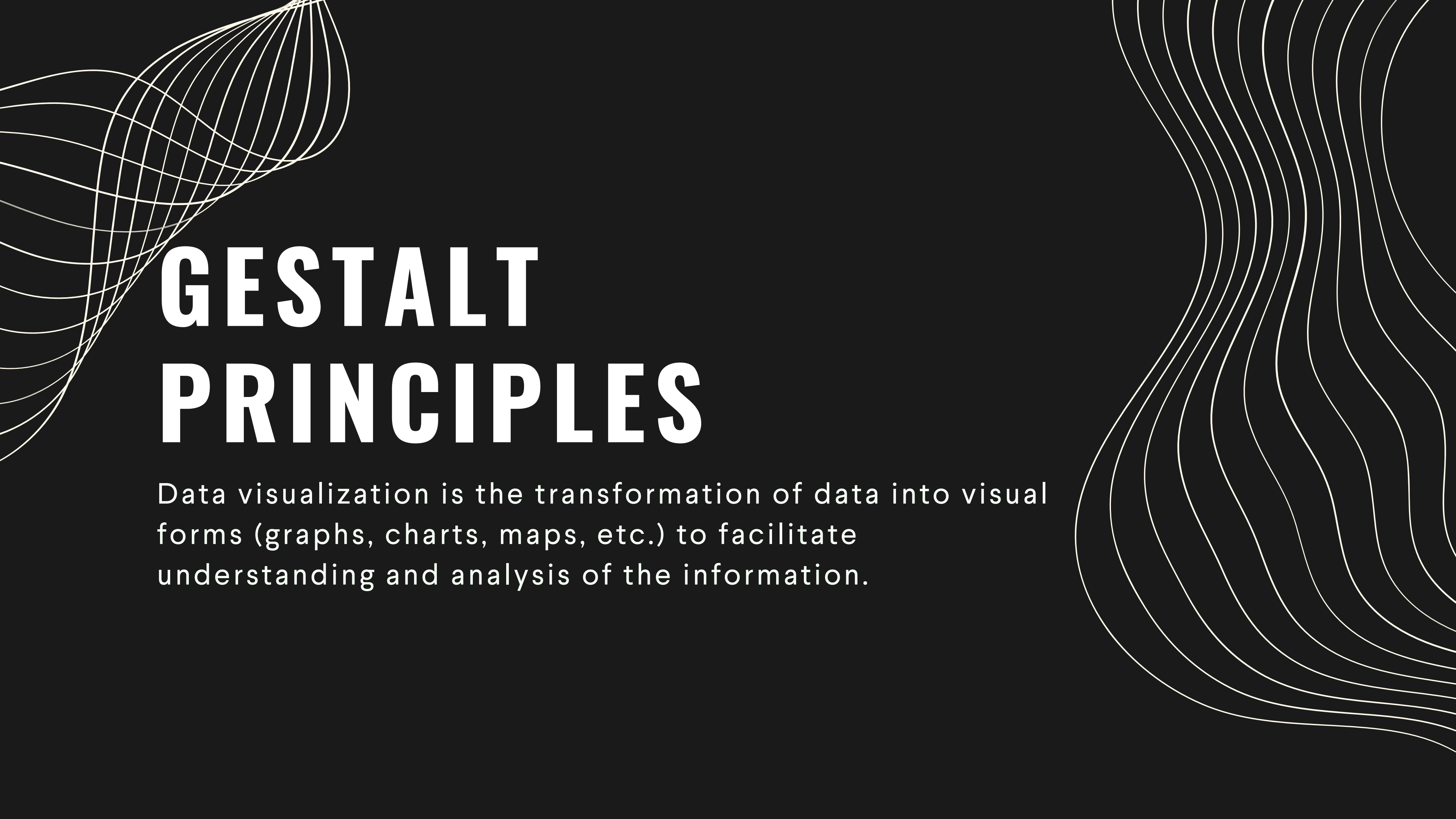
COLOR AND DATA INCONSISTENCY



■ YES
■ NO

BAD EXAMPLES

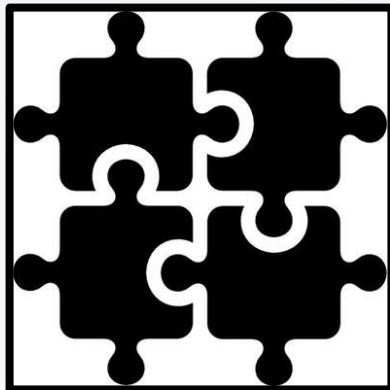




GESTALT PRINCIPLES

Data visualization is the transformation of data into visual forms (graphs, charts, maps, etc.) to facilitate understanding and analysis of the information.

GESTALT PRINCIPLES

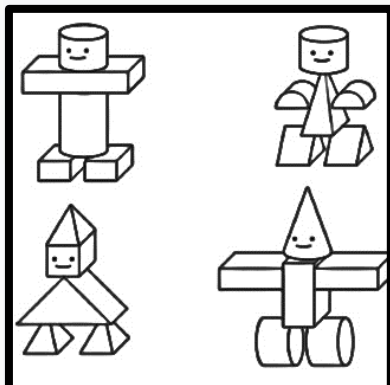


Emergence

First we see the form as a whole and only then individual details.

Reification

Our vision automatically fills in shapes even in the absence of details..



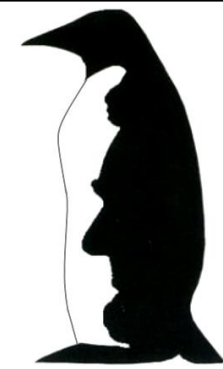
Invariance

We are able to recognize similar shapes that differ in shape, color, and location in space.

GESTALT PRINCIPLES

Multistability

The brain tries to process all possible interpretations of the form at once..



Background concept

We always try to divide the figure into foreground and background, separating the background, even if the figure is flat.

The influence of experience

Personal experience, environment, historical and cultural concepts influence our perception.



LIST OF BOOKS

